Michael Chen

DATA ENGINEER WITH 2+ YEARS IN PRODUCT & E-COMMERCE ANALYTICS

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Skills

Data Processing: Apache Spark (Core, Streaming), Kafka, Trino, Dataproc, Glue, Databricks
Data Storage & Modeling: BigQuery, Snowflake, Apache Iceberg, MySQL, PostgreSQL, dbt, Kimball Modeling, Star Schema
Orchestration & DevOps: Airflow, Docker, GCP, AWS, Git, Linux
Programming Languages: Python, SQL, Java

Experience

Merkle, *Data Engineer* — *June 2022 – February 2023*

- Engineered a cross-cloud ETL pipeline (BigQuery, AWS Glue, GCS, S3) processing ~100M Google Analytics events to support the biweekly retraining of L'Oréal's e-commerce recommendation system
- Designed a Kimball star schema over TB-scale dataset, partnering with machine learning engineers and data scientists to align on data requirements, enabling self-serve analytics and reducing query latency by 90%
- Optimized BigQuery partitioning and clustering strategies, reducing data scan volume by 80%; developed Looker Studio dashboards tracking recommendation algorithm A/B testing results for cross-functional stakeholders
- Implemented automated data quality checks with fallback backfill workflows, preventing empty recommendation outputs

Merkle, *Data Engineer Intern* — *March 2022 – June 2022*

- Automated end-to-end ML model refresh workflow on GCP (Vertex AI, Pub/Sub, Cloud Functions, Cloud Logging), eliminating manual query execution, model training, post-processing, and evaluation steps, reducing operational overhead by 85%

Freelance / Contract, *Data Engineer* — *June 2023 – August 2024*

- Eliminated manual document lookup for 6 industrial enterprise clients by building an end-to-end RAG pipeline powering QA chatbots, enabling instant retrieval over fragmented technical documentation
- Built a distributed text-to-embedding pipeline using Spark, leveraging mapPartitions to increase embedding throughput by 6x

Soochow University, *Research Assistant* — *October 2020 – June 2022*

- Engineered Python ETL pipelines to transform raw activity logs into curated feature tables, reducing data preparation time by 70% and ensuring data reproducibility for downstream ML analytics

Projects

Short Video Hybrid Analytics Lakehouse [Github] — *December 2025 – Present*

- Engineered a scalable analytics lakehouse (Kafka, Spark, Iceberg, Trino, Airflow) ingesting 5k+ events/sec across 3 Kafka topics and 7 event types to power retention and engagement analytics for short-video workloads
- Designed a medallion architecture ensuring reliable downstream analytics, flexible schema evolution, and efficient reprocessing
- Built dbt dimensional models aligned with business metrics, reducing Trino scan volumes and improving dashboard performance
- Mitigated Spark data skew via salting strategies to stabilize large-scale transformations; orchestrated daily batch workflows with Airflow, implementing DAG dependency management and SLA monitoring to meet d+1 analytics SLAs
- Developed Metabase dashboards visualizing retention and engagement KPI, enabling self-serve analytics for stakeholders

BookBridge: Recommendation Data Pipeline [Github] — *August 2025 – November 2025*

- Scaled a research recommender approach on GCP (GCS, Dataproc), building data pipelines to support 4M+ Amazon book items
- Engineered a PySpark ETL pipeline processing 30M+ ratings, mitigating data skew via broadcast joins to prevent OOM failures

Spatial Data Management System [Github] — *February 2025 – April 2025*

- Engineered a high-performance spatial database in Java, implementing a PR QuadTree index to optimize 2D range queries, reducing search time complexity from O(N) to O(log N), enabling sub-millisecond retrieval
- Implemented unit testing and performance benchmarking to validate query accuracy and ensure system stability under load

Education

Virginia Tech, *MEng in Computer Science* — *August 2024 – May 2026*

- **Relevant Courses:** Data Structures & Algorithms, Database Management Systems, Software Engineering, Machine Learning, Deep Learning, Natural Language Processing
- **Extracurricular Self-Study (Non-Degree):** Distributed Systems (MIT 6.824), Operating Systems (UC Berkeley CS162)
- **Competitive Programming:** LeetCode Contest Rating 1904 (Top ~4%)

Soochow University, *BBA in Big Data Management (Program renamed to Data Science)* — *September 2018 – June 2022*

- **Relevant Courses:** Data Structures & Algorithms, Big Data Systems, Big Data Analytics and Applications, Cloud Computing